

MODUL UNIVERSITY VIENNA

SEID 2364: Societal and Ethical Impact of Data Science

SESSION 5

Case Analysis and
structure of ethical
decisions

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Opening Dissections

Moving from Speculation to Research and Analysis

- Use Assignment 4
- Work precisely
- No generalization



(Write. Do not discuss yet.)

Dissection 1

Using your Assignment 4 case study, rewrite your use case as a decision.

- Actor → Decision → Target

No new information
No explanation or judgment

- Example: A bank decides whether to approve or deny a loan to an applicant.

(Write. Do not discuss yet.)

Dissection 2

What is NOT in your sentence?

List three elements NOT in your sentence that could change the outcome.

- Do not repeat elements already in your sentence
- Each element must affect the outcome

Rules

- Each element must be a single noun phrase (no full sentences)
- Do not repeat anything already in your decision sentence
- Each element must plausibly change the decision outcome (approve/deny, allow/block, rank/hide, etc.)

Example:

- Credit history
- Lending criteria
- Regulation

(Write. Do not discuss yet.)

Dissection 3

Determinants and Contributors

Identify:

- Determinants (what shapes outcome)
- Contributors (who participates)

Loan Example:

- Determinants:
- Bank Criteria
- Recipient credit data

Contributors:

- Bank
- Applicant
- Credit agency
- Policy

(Write. Do not discuss yet.)

Dissection 4

Apply two different ethical frameworks to your decision.

Instructions

- Identify the Frameworks you are using
- Evaluate possible outcomes
- Identify affected agents
- Compare harms and benefits
- Compare results across frameworks

Example

- Outcome A (approve): benefits applicant, risk to bank
- Outcome B (deny): protects bank, harm to applicant
- Frameworks may prioritize these differently.

(Write. Do not discuss yet.)

Dissection 4: Exemplar

Framework 1: **Consequentialism**

Outcome A (approve loan):

- Benefits: applicant gains access to capital
- Harms: increased financial risk to bank
- Affected agents: applicant, bank

Outcome B (deny loan):

- Benefits: reduced financial risk to bank
- Harms: applicant denied opportunity
- Affected agents: applicant, bank

Framework 2: **Deontology**

Outcome A (approve loan):

- Harms: violation of lending rules if criteria are not met
- Affected agents: bank, regulatory bodies
- Outcome B (deny loan):

- Benefits: adherence to established rules and duties
- Harms: applicant disadvantaged despite need
- Affected agents: applicant, bank

• **Comparison**

Consequentialism prioritizes overall outcomes and risk–benefit balance.

Deontology prioritizes rule compliance, even when outcomes differ.

(Write. Do not discuss yet.)

Dissection 5

Where frameworks disagree.

Identify **one use case** where applying

- **Framework A** leads to **Decision X**
- **Framework B** leads to **Decision Y**

Requirements

- Same decision context
- Same actors and target
- Different justified outcomes

Focus

- What assumption or priority changes?
- What harm is elevated or constrained?
- Why does the framework diverge here?

Where frameworks disagree.

Identify **one use case** where applying

- **Framework A** leads to **Decision X**
- **Framework B** leads to **Decision Y**

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- What harm is elevated or constrained?
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Dissection 5: Exemplar

Example: Same case, different outcomes

Decision

A bank decides whether to approve or deny a loan to an applicant.

***Consequentialism* → Approve**

The applicant gains access to capital, enabling housing stability or business growth.

The bank assumes some financial risk, but expected overall benefit (to applicant and possibly society) outweighs potential loss.

Harm is evaluated by aggregate outcomes and risk–benefit balance.

Recommended outcome: Approve the loan

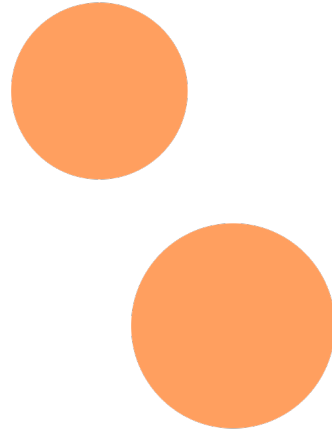
***Deontology* → Deny**

Lending rules and eligibility criteria are not fully met.

Approving would violate established duties and procedural fairness.

Even if approval would benefit the applicant, rule compliance takes priority.

Recommended outcome: Deny the loan



ZOTERO

ZOTERO CHECK

Confirm:

- Your Assignment 4 sources are uploaded
- Entries appear in your Zotero library

Next: We will review and discuss tagging.



Add a source to your Zotero library.

Why decisions differ

Consequentialism

This decision prioritizes **aggregate outcome and expected benefit**.

Approval is favored because the overall benefits (to the applicant and potential downstream effects) are judged to outweigh the financial risk to the bank.

Harm is evaluated by **net impact across affected agents**, not by rule compliance.

Deontology

This decision prioritizes **rule adherence and procedural obligation**.

Denial is favored because approving the loan would violate established lending criteria, regardless of beneficial outcomes.

Harm is defined as **breach of duty**, even when consequences appear favorable.



Evaluating Frameworks

Different results. Same system?

Focus: outcomes Source: Utilitarian tradition (Jeremy Bentham, John Stuart Mill)

Assessment Mechanism:

- Define outcome metrics (types of harm)
- Estimate probabilities and magnitudes - Aggregate (sum/expected value) across stakeholders - Sensitivity to weighting assumptions - Check for omitted harms (indirect, long-term, distributional)

PROBLEM

Frameworks are models, not reality

Models emphasize and omit

Different models → different conclusions

Disagreement is structural, not mistaken

Framework choice is the decision

No framework can justify itself

Comparison requires a meta-framework

A framework is a way to **model reality**, not reality itself.

Each framework models reality **differently**, emphasizing certain factors while excluding others.

As a result, different frameworks can produce **different evaluations of the same situation**.

This raises a problem: **how do we decide which framework is “best”?**

To answer that question, we must **step outside any single framework**.

We need a **higher-level framework** that allows us to compare multiple frameworks.

This larger framework must let us evaluate **how different models represent, prioritize, and resolve harm**.

Meta Frameworks

Different results. Same system?

If different frameworks provide different results, how do we develop a system that allows us to ethically evaluate the frameworks?

Is this even possible?

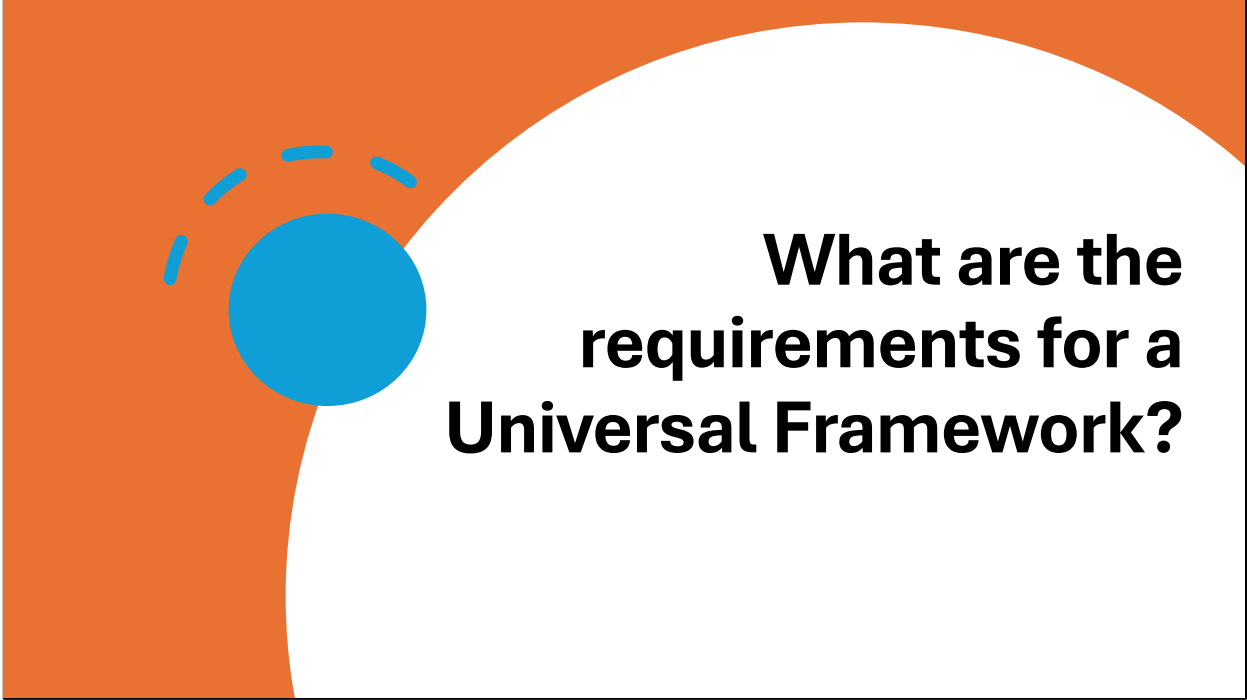


Meta-frameworks: stepping outside any single ethical lens to compare them.

Set up the puzzle: if Framework A and Framework B yield different verdicts on the same case, we cannot resolve the disagreement from inside either one.

Ask the room: is a neutral vantage point even possible, or does every meta-framework smuggle in its own commitments?

Goal of this section: motivate the search for evaluation criteria that frameworks themselves can be measured against.



What are the requirements for a Universal Framework?

Requirements for a Universal Framework — what would it have to satisfy?

Candidate criteria to discuss: internal consistency, generality across actors and contexts, impartiality, transparency of assumptions, and revisability in light of new evidence.

It must also handle pluralism: accommodating disagreement without collapsing into relativism.

Prompt the group to add or challenge criteria before we test any framework against them.



Can ethical behavior be applied to a tool?

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Distinguish moral agents (capable of intent and responsibility) from moral patients and from instruments.

Tools do not have intent, but their design encodes values and their use produces outcomes — so ethical evaluation attaches to designers, deployers, and users rather than to the tool itself.

Tee up the harder case: as tools become more autonomous (AI systems), where does that line blur, and what new responsibilities emerge?



Developing new ethical paradigms in combinative reality

Developing new ethical paradigms in combinative reality.

Combinative reality: human decisions and tool-mediated actions are increasingly entangled, so classical frameworks built around a single human agent strain to apply cleanly.

Discuss what a new paradigm might add: distributed responsibility, accountability across sociotechnical systems, and dynamic norms that update with capability.

Close by inviting the group to sketch what an ethics for hybrid human-tool action should look like — and what existing frameworks it can still borrow from.

CLOSING

Ethics depends on system description.

- Where we draw the boundary sets what counts as harm, who counts as an actor, and which outcomes we can even see.
- Change the system description and the verdict changes — not because the facts changed, but because a different set of facts came into view.
- No framework escapes this: each one presupposes a system — what is inside, what is outside, what is held fixed.
- The meta-question is not only “which framework is right?” but “which description are we reasoning over, and what does it leave out?”
- In a combinative reality of human and tool-mediated action, defining the system is itself an ethical act.
- **The commitment: make the system description explicit, test it against alternatives, and revise it as the world changes.**

How we draw the boundary determines what counts as harm, who counts as an actor, and which outcomes we are even able to see. Choose a different system description and the ethical verdict changes — not because the facts changed, but because a different set of facts came into view.

No single framework escapes this. Utilitarian, deontological, virtue, and rights-based lenses each presuppose a system — what is inside it, what is external, and what is held fixed. The meta-question of ethics, then, is not only “which framework is right?” but “which description of the system are we reasoning over, and what does that description leave out?”

In a combinative reality of human and tool-mediated action, this responsibility is sharper: defining the system is itself an ethical act. The honest closing move is not a final answer but a commitment — to make the system description explicit, to test it against alternatives, and to revise it as the world we are describing changes.